GGG-B-325B
July 18, 1980
SUPERSEDING
GGG-B-325A
June 15, 1972

FEDERAL SPECIFICATION

BINDER, LOAD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers lever-operated, eccentric load binders.

1.2 Classification.

1.2.1 Types, classes, and styles. Load binders covered by this specification shall be of the following types, classes, and styles as specified (see 6.2).

Type I - Plain.
Type II - With slack take-up hook arrangement.

Class 1 - Without compression spring.

Style A - Light duty.
Style C - Heavy duty.

Class 2 - With compression spring.

Type III - With continuous slack take-up double claw-hook arrangement.

Class 1 - Without compression spring.

Style A - Light duty.
Style B - Medium duty.
Style C - Heavy duty.

Class 2 - With compression spring.

Type IV - Latching type load binder with low energy release feature.

Class 1 - Without compression spring.

Style A - Light duty.
Style B - Heavy duty.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration, Business Service Centers in Boston; New York; Washington, DC; Philadelphia; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection for Attributes.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials (ASTM) Standards:

ASTM A 413 Carbon Steel Chain.

(Application for copies should be addressed to the American Society for Testing and Material, 1916 Race St., Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Materials of component parts of the binders shall be as specified in 3.1.1 through 3.1.7. The manufacturer shall utilize recovered materials to the maximum extent practicable.

3.1.1 Operating lever. The operating levers shall be of malleable iron, forged steel, or cast steel.

3.1.2 Swivels. The swivels shall be either malleable alloy or drop forged steel.

3.1.3 Hooks. The hooks shall be of high strength drop forged steel.

3.1.4 Clevis and tongue. The clevis and tongue shall be of forged steel or alloy steel.

3.1.5 Links. The links shall be of either alloy or forged steel.

3.1.6 Pins. The pins shall be of hardened steel.

3.1.7 Compression spring. The compression spring shall be of spring steel.

3.2 Illustrations. The illustrations shown herein are for the convenience of identification only and are not intended to preclude the purchase of load binders which are otherwise in accordance with the requirements of this specification.

3.3 Design. Types I and II load binders shall consist essentially of an operating lever, clevis, tongue, two swivels, two links, two pins, two grab hooks, and for type II, class 2, a compression spring. Type III load binders shall consist essentially of an operating lever, two clevises, a tongue, a swivel, three links, two claw hooks, a grab hook, and, for class 2, a compression spring. Type IV latching load binders shall consist essentially of an operating lever, front clevis, back clevis, two swivels, two links, two grab hooks, a handle pin, and a greased front clevis pin. The load binders shall be designed for eccentric takeup and latching type takeup. The load binders shall operate in accordance with the following:

(a) Provide a minimum takeup of 4 inches.
(b) Allow free movement of the assembled parts of the load binder.
(c) Preclude the possibility of the parts binding or spreading under normal service conditions.
(d) Withstand the minimum pulling load specified in 3.5, 3.6, and 3.7, as applicable, without breakage and permanent deformation.
(e) Provide suitable means for a positive and permanent lock whereby the lever will be held in a fixed position when the takeup operation is completed.
(f) Acceptable means for a positive and permanent lock (which may be provided includes, but is not restricted to, the following:

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(1) A pm fastened to the load binder by a chain and appropriate hole. in parts of the binder.
(2) A hook on a chain attached near the free end of the lever.
(3) Construction utilizing the slack end of the chain. (An over center pivot arrangement, by itself, is not considered a positive and permanent locking means. Also the chain slack end wrapped around the handle is not acceptable as a permanent locking means.)

3.3.1 Operating lever. The operating lever shall be of such design as to have a tie bar or solid back construction.

3.4 Interchangeability of parts. All parts of a particular model load binder shall be interchangeable with the similar parts of any other load binder of the same model made by the same manufacturer.

3.5 Type I, plain load binder. Type I load binders shall be capable of binding any load that requires the use of 1/4-inch up to and including 1/2-inch chain. The length of the operating lever shall be not more than 18-1/2 inches. The load binders shall be capable of withstanding a tensile proof load of 16,000 lb (see 4.3.1.1). The load birders shall weigh not less than 8lb nor more than 14 Jb and shall be similar to figure I.

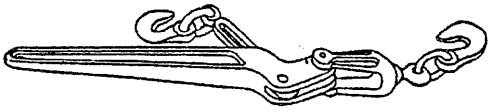


FIGURE I - TypeI I, plain

3.6 Type II, load binders with slack takeup hook. Type II load binders shall be provided with a slack takeup hook arrangement and shall be constructed in such a manner that tension will be maintained on the chain at all times during the takeup operation. Class 1, style A load binders shall have an operating lever not more than 18 inches in length and shall be capable of withstanding a tensile proof load of 12,000 lb (see 4.3.1.2.1). Class 1, style C and class 2 load binders shall have operating levers not more than 20 inches in length and shall be capable of withstanding a tensile proof load of 15,000 lb (see 4.3.1.2.2).

3.6.1 Slack takeup hook arrangement. The slack takeup hook arrangement of type 11 load binders shall be simple and easy to operate and shall have a leverage of not more than a IO-inch movement at the end of the operating lever to 1-inch movement of the slack takeup hook. The slack takeup hook of class 1, style A load binders shall be capable of withstanding a proof load of 6,000 lb (see 4.3.2.1.1). The slack takeup hook of class 1, style C, and class 2 load binders shall be capable of withstanding a proof load of 8,000 lb (see 4.3.2.1.2 and 4.3.2.2).

3.6.2 Class 1, load binders without compression spring. Type II, class 1, styles A and C load binders shall be similar to figure 2.

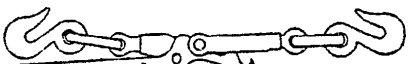


FIGURE 2 - Type II, class 1, styles A and C, with slack takeup hook arrangement, without coinpression spring

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3.6.21 Style A, light duty. Type II, class 1, style A load binders shall be provided with grab hooks and a slack takup hook. The load binders shall be designed for use with 1/4- and 3/8-inch chains. The chain shall meet ASTM A 413, table 3, link dimensions of Proof Coil (PC) and High Test (HT) lid and chain for both 0.281 (1/4-inch) and 0.406 (3/8-inch) link diameter.

3.6.2.2 Style C, heavy duty. Type II, class 1, style C load binders shall be provided with grab hooks and a slack takeup hook. The load binders shall be designed for use with 3/8- and l/2-inch chains.

3.6.3 Class 2, load binders with compressioning. Type II, class 2 load binders shall be similar to figure 3. The load binders shall be provided with grab hooks and a slack takeup hook designed for the use with 3/8- and 1/2inch chains. The load binder shall be provided with a compression spring assembly which shall function as a link.

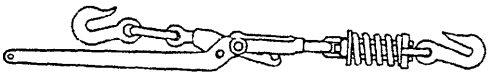


FIGURE 3- Type II, class 2,
with slack takeup hook
arrangement, with compression spring

3.6.3.1 Compression spring. The compression spring shall be capable of exerting a force of 5,000 lb ± 200 lb immediately prior to becoming solid without taking a permanent set. The spring shall have a mean free diamiter of not more than 3-1/2 inches and shall have a rate not greater than 4,000 lb per inch deflection. Hook or hairpin loops external to the spring will not be acceptable.

3.7 Type III, load binders with continuous slack takeup double claw hook. Type 111 load binders shall be provided with continuous slack takeup double claw hook arrangement and be constructed in such a manner that tension will be maintained on the chain at all times during the takeup operation. Type III load binders shall have an operating lever not more than 22 inches in length. Class 1, style A, and class 2 load binders shall be capable of withstanding a tensile proof load of 7,000 lb (see 4.3.1.4.1) Class 1, style B load binders shall be capable of withstanding a tensile proof load of 19,000 lb (see 4.3.1.4.2). Class I, style C load binders shall be capable of withstanding a tensile proof load of 28,000 lb (see 4.3.1.4.3).

3.7.) Continuous slack takeup double claw hook arrangement. The continuous slack takeup double claw hook arrangement of type III load binders shall be easy to engage and lock onto the chain and shall have a leverage of not more than 10-inch movement at the end of the operating lever to 1-inch movement of the claw hook engaged on the chain.

3.7.2 Class 1, load binders without compression spring. Type III, class 1, styles A, B, and C load binders shall be similar to figure 4.

3.7.2.1 Style A, light duty. Type III, class 1, style A load binders shall be designed for use with 3/16- and 1/4-inch chains.

3.7.2.2 Style B, medium duty. Type III, class 1, style B load binders shall be designed for use with 5/16- and 3/8-inch chains.

3.7.2.3 Style C, heavy duty. Type III, class 1, style C load binders shall be designed for use with 7/16- and 1/2 inch chains.

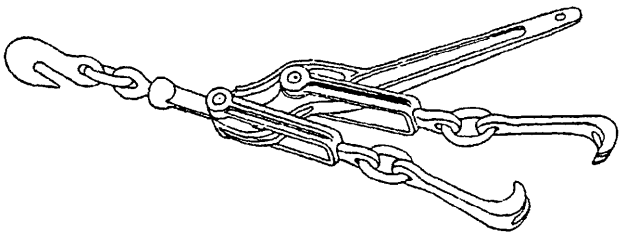


FIGURE 4- Type III, class 1, styles A,
B, and C, with continuous slack takeup
double claw hook arrangement, without
compression spring

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3.7.3 Class 2, load binders compression spring. Type III, class 2 load binders shall be similar to figure 4 except that, the grab hook assembly shall be provided with a compression spring assembly which shall function as a link (see figure 3). The load binders shall be designed for use with 3/8- and 1/2-inch chains. The compression spring shall conform to the requirements of 3.6.3.1.

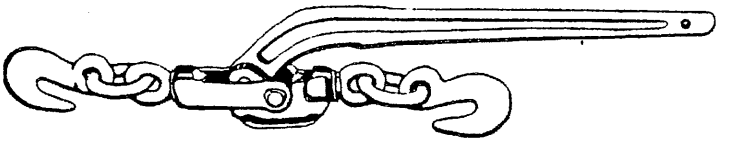


FIGURE 5 -Type IV, latching load binder.

3.7.4 Type IV, latching load binder. Type IV latching load binders shall be provided with a handle and sliding front clevis pin so that energy can be transferred to a back clevis for low energy release feature. Type IV load binders shall have an operating lever of not more than 19 inches in length. Class 1, style A load binders shall be capable of withstanding a tensile proof load of 12,000 lb (see 4.3.2.3.1) and class 1, style C, a load of 15,000 lb (see 4.3.2.3.2). The load binders shall weigh not less than 8 lb nor more than 14 lb and shall be similar to figure 5.

3.7.4.1 Style A, light duty. Type IV, class 1, style A load binders shall be designed for use with 1/4- through 3/8-inch chain sizes.

3.7.4.2 Style C, heavy duty. Type IV, class 1, style C load binders shall be designed for user with 3/8- through 1/2-inch chain sizes.

3.8 Finish. The load binders shall be free from fins, burrs, and excessive roughness and shall have a suitable protective coating or be painted. When painted, the color shall be in accordance with the manufacturer's standard practice.

3.9 Identification marking. Each load binder shall be plainly marked with the manufacturer's designation for the item and with his name or trademark of such known character that the source of manufacture may be readily determined.

3.10 Workmanship. The load binders shall be free from defects which may detract from their appearance or affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purt-base order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Inspection for acceptance. Inspection shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that all components and materials used were manufactured, tested, and inspected in accordance with the requirements or referenced subsidiary specifications and standards to the extent specified herein or, if none, in accordahce with this specification.

4.2.2 End product inspection

4.2.2.1 Visual examination. Representative samples of the completely assembled end item shall be examined for defects in finish, material, construction, workmanship and marking. Defects shall be classified as outlined in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in defects per hundred units.

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Examine	Defect	Classification Major	Minor
Finish, protective	Any breaks through coating to metal.		X
	Any area of rust or corrosion.	X	
	Dirt, oil, grease, or other foreign matter.		X
Materials	Component not fabricated of the specified material.	X	
Design	Any characteristic not in accordance with the specified requirements.	X	
Construction and Workmanship	Any component cracked, fractured, or otherwise damaged affecting serviceability.	X	
	Defective or other damaged, but not affecting serviceability.		X
	Any component missing.	X	
	Any operation omitted or not performed as specified	X	
	Any sharp edges, butts, or metal slivers.	X	
	Functioning component that requires abnormal force to operate.		X
	Any part not assembled or joined as specified.	X	
Identification marking	Missing, incomplete, not legible, or incorrect.		X

4.2.2.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. The inspection level shall be S-2 with an AQL of 6.5 defects per hundred units.

4.2.2.3 Sampling for tests. A random sample of load binders shall be taken from each inspection lot in accordance with MIL-STD-105 at inspection level S-3 for tests of 4.3. If any load binder fails any of the tests, the lot shall be rejected.

4.3 Tst procedures.

4.3.1 Tensile load test of load binders.. Sample load binders shall be tested in any suitable tensile testing device. The binder shall be secured in the testing device by its grab hooks. The operating handle of the binder shall be in the closed or load binding position during the test. The binders shall be loaded as specified in 4.3.1.1, 4.3.1.2.2, 4.3.1.3, 4.3.1.4.2, and 4.3.1.4.3, following each loading test, the binder shall be examined for failure of any component and permanent deformation of the total binder.

4.3.1.1 Type I, load binders. Type I oad binders shall be tension loaded to 16,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

4.3.1.2 TYPe II, class I, load binders.

4.3.1.2.1 Style A. Type II, class 1, style A load binders shall be tension loaded to 12,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

4.3.1.2.2 Style C. Type II, class i, style C load binders shall be tension loaded to 15,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

4.3.1.3 Type II, class 2, load binders. The test procedure and loading shall be identical with that for type II, class 1, style C, as specified in 4.3.1.2.2.

4.3.1.4 Type III, class 1, load binders.

4.3.1.4.1 Style A. Type III, class 1, style A, and class 2 load binders shall be tension loaded to 7,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

4.3.1.4.2 Style B. Type III, class 1, style B load binders shall be tension loaded to 19,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

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4.3.1.4.3 Style C. Type III, class 1, style C load binders shall be tension loaded to 28,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

4.3.2 Slack takeup hook load test. Sample load binders shall be tested in any suitable tensile testing device to determine conformance to the slack takeup hook loading requirements of 3.6.1. The binder shall be secured in the testing device by the trailing grab hook and the slack take-up hook. The shank of the slack take-up hook shall be vertical to a tongitudinal line through the grab hooks during the test. The takeup hook and components of the binder that were subjected to loading shall be examined for any failure and permanent deformation.

4.3.2.1 Type II, class 1, load binders.

4.3.2.1.1 Style A. Type II, class 1, style A load binders shall be loaded as specified in 4.3.2 to 6,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

4.3.2.2 Style C. Type II, class 1, style C load binders shall be loaded as specified in 4.3.2 to 8,000 lb, this load being applied for not less than 1/2 minute. The load shall then be removed and the load binder examined.

4.3.2.2 Type II, Class 2, load binders. The test procedure and loading shall be identical with that for type II, class 1, style C load binders as specified in 4.3.2.1.2.

4.3.2.3 Type IV, load binders.

4.3.2.3.1 Style A. Class 1 load binders shall be tension loaded to 12,000 lb with load applied for not less than 1/2 minute. The load shall be removed and the load binder examined for any cracks or other damage that will cause the binder to fail.

4.3.2.3.2 Style C. Class 1 load binders shall be tension loaded to 15,000 lb with load applied for not less than 1/2 minute. The load shall be removed and the load binder examined for any cracks or other damage that will cause the binder to fail.

4.3.3 Compression spring test. Sample compression springs for class 2 load binders shall be tested in any suitable testing device to determine conformance to the requirements of 3.6.3.1 or 3.7.3.1 as applicable. The spring shall be compressed to a length immediately prior to its becoming solid and shall be held in this position for not less than 1/2 minute. The load shall then be removed and the spring examined for any failure and permanent deformation.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery and selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method or application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of comtainer flaps, loose strapping, and inadequate strapping, and inadequate stapling.
	Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The packaging, packing shall be level A or commercial as specified in 6.2.

5.1.1 Level A. The load binder shall be packaged, packed, and marked in accordance with PPP-P-40.

5.1.2 Commercial. The binders shall be packaged in accordance with normal commercial practice and packed in a manner that will assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with National Motor Freight Classification and Uniform Freight Classification.

5.1.2.1 Marking shall be in accordance with the contract or purchase order.

6. NOTES

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6.1 Intended use. The load binders covered by this specification are intended for use as a tiedown tensioning device for binding such loading as logs, lumber, pipes, beams, hay, and other materials on open trucks, fiat cars, and ships.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents;

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2.1).
(c) Level of packaging required (see 5.1).
(d) Level of packing required (see 5.1).
(e) Additional marking, when required (see 5.1).

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